

SMART URBAN MOBILITY AND TRAFFIC INTELLIGENCE

Power BI Dashboard Project Documentation

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Project Information	Details
Project Type	Data Analytics
Tool	Microsoft Power BI Desktop
Dataset	Smart City Transportation
Main Dashboard	Smart Urban Mobility and Traffic Intelligence
Key Techniques	Power Query, Data Modeling, DAX, Calculated Column, KPI Cards, Slicers and Charts

1. Project Overview

Smart Urban Mobility and Traffic Intelligence is an interactive Power BI dashboard developed to analyze urban transportation data. The project converts raw trip-level information into meaningful KPIs and visual insights that can be explored using filters.

The dashboard focuses on transportation activity, passenger volume, traffic density, vehicle speed, travel time, transport modes, vehicle types and city-level trip distribution. The report is designed to give a quick overall view while also allowing detailed analysis through slicers.

2. Problem Statement

Urban mobility datasets contain many records and measures, making it difficult to identify important patterns directly from raw data. A decision-oriented dashboard is required to summarize transportation performance and allow users to compare cities, modes and vehicle types efficiently.

3. Project Objectives

- Understand and prepare the smart-city transportation dataset.
- Create useful DAX measures for dynamic KPIs.
- Add a Population Density calculated column for extended urban analysis.
- Build KPI cards for major transportation indicators.
- Add City, Transport Mode and Year slicers.
- Create charts for city, transport mode, vehicle type and travel-time/speed analysis.
- Develop a clean, consistent and professional Power BI dashboard.

4. Dataset Understanding

The dataset contains transportation and urban-mobility information. The reference dataset contains approximately 500 rows and 32 columns. The main Power BI table used for the dashboard is Smartcitydata, with PopulationData available for population-related analysis.

Field	Use in Dashboard
Trip_ID	Identifies trips and supports trip counting.
City	City comparison and City slicer.
Transport_Mode	Private, Public and Shared analysis.
Vehicle_Type	Bus, Taxi, Car, EV and Motorcycle analysis.
Passenger_Count	Total passenger KPI.

Traffic_Density	Average traffic-density KPI.
Average_Speed_KMH	Average speed KPI and scatter chart.
Travel_Time_Min	Average travel-time KPI and scatter chart.
Year	Year slicer.
Population / Area	Used for Population Density analysis.

5. Data Integration and Preparation

The transportation data was imported into Power BI Desktop and reviewed before building the report. Relevant numerical and categorical fields were identified for calculations, filtering and visualization.

- Checked numerical and categorical data types.
- Reviewed City, Transport_Mode and Vehicle_Type values.
- Verified Trip_ID, Passenger_Count, Traffic_Density, Average_Speed_KMH and Travel_Time_Min.
- Checked Year values used in the report.
- Validated the prepared data before visualization.

Trip_ID	Vehicle_ID	Vehicle_Type	Transport_Mode	Route_ID	City	Region	Latitude	Longitude	Smart_Zone_ID	Traffic_Density	Average_Speed_KMH	Congestion_Level
TRP0000002	VEH49966	Car	Shared	R4988	Tokyo	Central	-20.46082	-131.59387	SZ0028	0.79	62.96	Medium
TRP0000005	VEH34821	Car	Public	R4311	Mumbai	Central	-25.93007	-155.26735	SZ0396	62.53	61.13	Low
TRP0000018	VEH43467	Motorcycle	Private	R0366	London	Central	74.20079	-20.98353	SZ0179	47.28	21.6	High
TRP0000022	VEH07657	Car	Shared	R3249	Mumbai	Central	-87.59407	-48.55506	SZ0751	30.68	23.82	Medium
TRP0000026	VEH40122	Taxi	Public	R1996	New York	Central	5.44043	-95.08639	SZ0486	18.76	96.42	Medium
TRP0000028	VEH37950	Bus	Public	R0372	London	Central	-0.63712	-146.63502	SZ0780	86.16	117.86	Medium
TRP0000030	VEH36217	Taxi	Public	R2878	Tokyo	Central	-79.6177	142.91546	SZ0047	61.35	117.31	Low
TRP0000035	VEH06985	Car	Shared	R2864	Tokyo	Central	-1.09784	176.31002	SZ0477	33.8	116.42	Medium
TRP0000039	VEH16383	EV	Private	R3861	Singapore	Central	81.79647	152.65988	SZ0718	11.3	65.68	Low
TRP0000040	VEH04220	Car	Private	R2661	Singapore	Central	76.07832	73.33847	SZ0885	84.02	99.86	Medium
TRP0000041	VEH34331	Car	Public	R4715	London	Central	-80.58847	-20.12412	SZ0525	5.13	30.09	Low
TRP0000047	VEH19529	EV	Public	R0677	New York	Central	-80.57048	122.76274	SZ0519	44.57	40.41	Severe
TRP0000048	VEH05085	Bus	Private	R4655	London	Central	28.751	-89.74922	SZ0251	35.83	11.76	Severe
TRP0000050	VEH15506	Motorcycle	Public	R2936	New York	Central	51.78984	117.22231	SZ0710	47.35	76	Low
TRP0000056	VEH19299	Car	Private	R1741	London	Central	-29.71294	-79.36132	SZ0293	99.66	100.77	Severe
TRP0000062	VEH34825	Taxi	Private	R2787	London	Central	18.95676	170.66579	SZ0811	43.81	36.54	High
TRP0000065	VEH40542	Taxi	Public	R4601	Singapore	Central	83.92599	89.07975	SZ0436	39.94	17.56	High
TRP0000069	VEH43655	Bus	Shared	R4565	Mumbai	Central	-36.04819	35.2332	SZ0839	58.36	32.18	High
TRP0000075	VEH40460	Taxi	Shared	R4952	London	Central	-10.75072	82.36954	SZ0355	66.16	102.91	Medium
TRP0000077	VEH24569	Taxi	Public	R4942	Tokyo	Central	60.41594	42.08644	SZ0514	7.9	117.18	Severe
TRP0000078	VEH38531	EV	Shared	R1883	Tokyo	Central	16.08288	-148.30993	SZ0510	75.52	57.6	Low
TRP0000080	VEH48407	Taxi	Public	R4599	Tokyo	Central	41.85808	114.03529	SZ0332	89.98	16.2	Medium
TRP0000083	VEH07525	Taxi	Private	R3455	Singapore	Central	-81.28683	-56.1273	SZ0829	36.36	22.19	Medium
TRP0000086	VEH18224	Motorcycle	Private	R2622	Singapore	Central	-15.21965	-129.04456	SZ0379	4.81	62.29	Severe
TRP0000100	VEH38144	Bus	Private	R1902	Singapore	Central	-4.04816	42.45681	SZ0717	70.51	44.2	Medium
TRP0000101	VEH34725	EV	Public	R1066	Mumbai	Central	-54.31783	136.368	SZ0538	43.23	110.37	Low
TRP0000102	VEH43754	EV	Private	R4763	Mumbai	Central	-41.92399	-173.38416	SZ0676	3.55	111.16	High

6. Data Modeling

The Power BI model uses Smartcitydata as the primary transportation table and PopulationData as a supporting population table. The model is organized around transportation measures and descriptive fields used for grouping and filtering.

Model Component	Purpose
Smartcitydata	Main transportation/trip-level data.
PopulationData	Population-related information.
City	Geographical analysis and filtering.
Transport_Mode	Transport category filtering.
Vehicle_Type	Vehicle category analysis.
Year	Time-based filtering.

7. Population Density Calculated Column

A Population Density calculated column was created as an additional analytical field. Population density helps connect transportation activity with the concentration of population in an urban area.

Population Density = Population ÷ Area

The calculated column can be used later for city comparisons, maps and analysis of the relationship between population concentration and transportation demand.

City	Population	Area_km2	Population_Density
Mumbai	12691836	603.4	21033.86808
Tokyo	14047594	2194	6402.731996
New York	8097282	778.2	10405.14264
Singapore	6036860	719.1	8395.021555
London	8945309	1572	5690.400127

8. DAX Measures

DAX measures were created so that KPI values respond dynamically to slicer selections and the current filter context. This makes the dashboard interactive rather than dependent on fixed values.

8.1 Total Passengers

Total Passengers = SUM(Smartcitydata[Passenger_Count])

Calculates the total number of passengers for the current filter selection.

8.2 Total Trips

Total Trips = COUNT(Smartcitydata[Trip_ID])

Counts the recorded trips in the current filter context

8.3 Average Traffic Density

Average Traffic Density = AVERAGE(Smartcitydata[Traffic_Density])

8.4 Average Speed

Average Speed = AVERAGE(Smartcitydata[Average_Speed_KMH])

8.5 Average Travel Time

Average Travel Time = AVERAGE(Smartcitydata[Travel_Time_Min])

9. KPI Cards

Five KPI cards were added to the top of the dashboard to provide an immediate summary of the transportation dataset.

KPI	Purpose
Average Traffic Density	Shows the average traffic-density level.
Total Passengers	Shows total passenger volume.
Total Trips	Shows total recorded trips.
Average Speed (km/h)	Shows average vehicle speed.
Average Travel Time	Shows average travel time in minutes.

In the overall unfiltered view, the dashboard displays approximately 49.91 average traffic density, 25K passengers, 500 trips, 64.50 km/h average speed and 91 minutes average travel time.

10. Slicers and Interactivity

Three slicers were added to allow users to interactively filter the dashboard.

Slicer	Values / Purpose
City	Filters transportation activity by city.
Transport_Mode	Filters Private, Public and Shared transport.
Year	Filters the dashboard by year such as 2024 and 2025.

The slicers update the KPI cards and related visuals, allowing the same report page to be used for both overall and focused analysis.

11. Charts and Visualizations

11.1 Trips by Transport Mode

A **pie chart** compares the distribution of trips among Private, Public and Shared transportation modes.

11.2 Trips by City

A **column chart** compares trip counts across cities. In the overall view, Singapore records 110 trips, followed by London with 103, Tokyo with 102, New York with 94 and Mumbai with 91.

11.3 Trips by Vehicle Type

A **donut chart** compares trips across Bus, Taxi, Car, EV and Motorcycle categories.

11.4 Travel Time vs Speed

A **scatter chart** compares Average Speed (KM/H) on the X-axis with Travel Time (Min) on the Y-axis. It helps visualize variation and possible relationships between speed and travel time.

Chart Design Changes

- Replaced default chart titles with meaningful titles.
- Enabled and formatted data labels where appropriate.
- Used consistent dark visual containers and light text.
- Improved chart spacing and alignment.

- Used readable category labels and legends.

12. Dashboard Design and Changes Made

The final dashboard uses a light-blue canvas with dark navy visual containers to create a consistent professional appearance. The layout follows a simple top-to-bottom analytical flow.

Dashboard Area	Content
Header	SMART URBAN MOBILITY AND TRAFFIC INTELLIGENCE
Top Row	Five KPI cards and three slicers
Middle Row	Trips by Transport Mode, Trips by City, Trips by Vehicle Type
Bottom Row	Travel Time vs Speed scatter chart

Main Changes Implemented

- Added a professional project title.
- Created five KPI cards.
- Added City, Transport_Mode and Year slicers.
- Formatted chart titles and data labels.
- Added the Population Density calculated column.
- Created DAX measures for dynamic KPI calculations.
- Applied a consistent dark-blue visual theme.
- Improved visual positioning, spacing and readability.
- Enabled interactive filtering across the dashboard.

Final Dashboard



Milestone 2

1. Project Overview

Smart Urban Mobility and Traffic Intelligence is a Power BI-based analytics project designed to understand urban transportation performance from trip, passenger, route, traffic and vehicle data. Milestone 2 focuses on converting the prepared dataset into an interactive dashboard that provides quick operational insights through KPI cards, slicers, charts and a geographical map.

The dashboard provides a consolidated view of total trips, total passengers, total disruptions, reliability, average speed, signal delay, route performance and performance over time. It also allows users to filter the analysis by city, transport mode and year.

2. Objectives

- Develop an interactive Power BI dashboard for smart urban mobility analysis.
- Analyze transportation demand through trips and passenger volume.
- Evaluate traffic and operational performance using speed, travel time, traffic density and signal delay.
- Compare performance across cities and routes.
- Use latitude and longitude to visualize traffic information geographically.
- Create dynamic DAX measures for important KPIs.
- Add slicers for City, Transport Mode and Year.
- Provide a clear and professional dashboard suitable for decision-making.
- Prepare the dashboard for deeper drill-down or drill-through analysis.

3. Dataset Understanding

The transportation dataset contains 500 records and multiple fields describing mobility, traffic, routes, vehicles and environmental conditions. The fields selected for Milestone 2 were reviewed to identify measures, categories and geographical attributes required for dashboard development.

- Trip and demand fields: Trip_ID, Passenger_Count and related trip information.
- Route fields: Route_ID and Smart_Zone_ID.
- Geographical fields: City, Region, Latitude and Longitude.
- Traffic fields: Traffic_Density, Signal_Delay_Sec and Road_Condition_Score.
- Performance fields: Average_Speed_KMH and Travel_Time_Min.
- Category fields: Transport_Mode, Vehicle_Type and Weather_Condition.
- Time fields: Timestamp, Month, Quarter and Year.

A PopulationData table was also used to support population-related analysis, including the Population_Density field.

4. Data Cleaning & Transformation

The dataset was prepared in Power Query before creating the final dashboard. Data types and field values were checked so that calculations, slicers and visualizations would work correctly.

- Checked and corrected numerical, text and date/time data types.
- Reviewed City, Route_ID, Transport_Mode and Vehicle_Type values.
- Prepared Latitude and Longitude as decimal-number fields.
- Validated Traffic_Density, Average_Speed_KMH, Travel_Time_Min and Signal_Delay_Sec.
- Removed unnecessary fields where appropriate.
- Validated the transformed dataset before loading it into the Power BI model.

Integrated population information and used Population_Density as an additional analytical field

5. Latitude and Longitude Preparation

For the geographical map, city-based coordinates were used so that the five cities in the dashboard appear at their correct locations. The coordinates were stored as decimal-number fields and assigned to the corresponding city records.

- London – 51.5074, -0.1278
- Mumbai – 19.0760, 72.8777
- New York – 40.7128, -74.0060
- Singapore – 1.3521, 103.8198
- Tokyo – 35.6762, 139.6503

6. DAX Measures

DAX measures were created to make the dashboard dynamic. Unlike fixed values, measures respond to the current filter context. Therefore, selecting a city, transport mode or year updates the relevant KPI values and visuals.

- Total Trips – counts the recorded Trip_ID values.
- Total Passengers – calculates total Passenger_Count.
- Total Disruptions – calculates the available disruption records.
- Reliability % – calculates the percentage of trips with no signal delay compared with total trips.

- Average Speed – calculates the average Average_Speed_KMH.
- Average Travel Time – calculates the average Travel_Time_Min.

7. Reliability KPI

The Reliability % measure was formatted as a percentage with two decimal places. This KPI is intended to provide a quick indication of trips without recorded signal delay and changes when the dashboard is filtered.

8. KPI Cards

KPI cards are positioned at the top of the dashboard to give users an immediate summary of the main transportation indicators.

- Total Trips – overall recorded trip count.
- Total Passengers – total passenger volume.
- Total Disruptions – disruption count available in the dataset.
- Reliability % – calculated reliability indicator.
- Average Speed – average transportation speed in km/h.

9. Slicers

Slicers provide interactive filtering and allow users to focus the dashboard on a particular city, transport mode or year.

- City: London, Mumbai, New York, Singapore and Tokyo.
- Transport Mode: Private, Public and Shared.
- Year: 2024 and 2025.

When a slicer is selected, the DAX measures and connected visuals update according to the selected filter context.

10. Charts

10.1 Signal Delay by City

The horizontal bar chart compares signal delay across the five cities. It helps identify cities with relatively higher or lower average signal-delay values.

10.2 Performance Over Time

The line chart presents average speed by month. It helps identify performance changes, peaks and declines during the year.

10.3 Route-wise Performance

The route-wise horizontal bar chart compares average travel time by Route_ID. This provides a route-level view of transportation performance and helps identify routes for further investigation.

10.4 Geographical Traffic Density

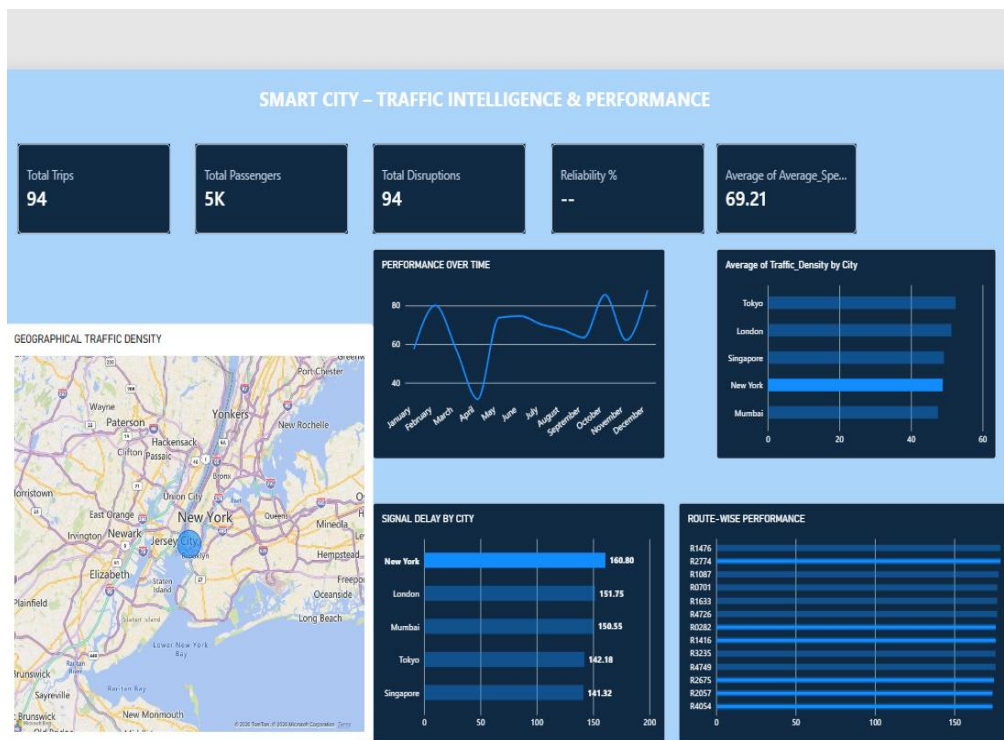
The map visual displays traffic-density information geographically using the corrected Latitude and Longitude values. Traffic_Density is used to control the bubble size, allowing higher-density locations to be visually distinguished.

11. Latitude/Longitude Map

A normal Power BI Map visual was added to provide geographical analysis. The map uses the prepared coordinates for London, Mumbai, New York, Singapore and Tokyo. This improves the dashboard by connecting numerical traffic information with actual geographic locations.

- Latitude → Latitude
- Longitude → Longitude
- Bubble Size → Traffic_Density
- Tooltips → Route_ID and relevant performance fields
- City → used to identify the geographical location

13. Dashboard Design



The dashboard follows a consistent professional design using a light-blue background, dark-blue visual

panels and blue chart elements. The layout is organized so that users can move from high-level KPIs to detailed geographical, city, route and time-based analysis.

- Clear title: SMART CITY – TRAFFIC INTELLIGENCE & PERFORMANCE.
- KPI cards placed across the top for quick understanding.
- Map used for geographical traffic-density analysis.
- Line chart used for performance over time.
- Bar charts used for city signal delay and route-wise performance.
- Consistent colors and spacing maintained across the dashboard.
- Slicers can be used to interactively filter the report.

13. Milestone 2 Deliverables

- Project objectives and dataset understanding documented.
- Data cleaning and transformation completed in Power Query.
- Population_Density field included for supporting analysis.
- DAX measures created for key transportation KPIs.
- KPI cards implemented.
- City, Transport Mode and Year slicers implemented.
- Route-wise Performance chart implemented.
- Signal Delay by City chart implemented.
- Performance Over Time line chart implemented.
- Latitude/Longitude geographical traffic-density map implemented.
- Professional dashboard design applied.
- Dashboard prepared for deeper drill-through analysis.

MILESTONE 3 – DEMAND INTELLIGENCE & MOBILITY ANALYTICS

1.1 Objectives

- Develop a demand-intelligence dashboard for urban mobility and passenger activity.
- Analyze ride-hailing trips across major cities.
- Compare travel time and transport-mode usage between cities.
- Measure micromobility usage by city.
- Study passenger demand under different weather conditions.
- Analyze rainfall intensity against passenger demand.
- Provide clear KPIs and visuals for mobility analysis.

1.2 Dataset Understanding

The Smartcitydata dataset contains trip-level mobility information. The selected fields support passenger-demand, transportation, location, performance, traffic and weather analysis.

Category	Important Columns	Purpose
Demand	Passenger_Count, Trip_ID	Measure passenger activity and trips
Mobility	Transport_Mode, Vehicle_Type, Route_ID	Analyze mobility choices
Location	City, Region, Latitude, Longitude	Compare geographic patterns
Performance	Average_Speed_KMH, Travel_Time_Min, Signal_Delay_Sec	Evaluate travel efficiency
Weather	Rainfall_MM, Weather_Condition, Temperature_C	Study weather sensitivity
Traffic	Traffic_Density, Congestion_Level	Understand traffic conditions

1.3 Data Preparation

- Numeric fields were retained as numeric types for aggregation.
- Categorical fields were used for city, transport mode and weather comparisons.
- Rainfall_MM was grouped into Low, Moderate, High and Very High categories.
- Rainfall Order was created from Rainfall_MM so categories appear in logical order.

2.1 KPI Cards

Four KPI cards provide a quick summary of the mobility dataset.

KPI	Displayed Value	Purpose
Average Speed	116.70	Average recorded travel speed
Average Travel Time	145.00	Average travel duration

Total Passengers	98	Passenger-demand measure
Average Traffic Density	41.98	Average traffic-density level

2.2 Ride-Hailing and City Analysis

- Ride-Hailing Trips by City compares activity across Singapore, London, Tokyo, New York and Mumbai.
- Average Ride Time by City compares travel duration across the five cities.
- Micromobility Usage by City highlights differences in micromobility activity.

2.3 Transport Mode Analysis

- Transport Mode Mix by City compares Private, Public and Shared transportation.
- Trips by Transport Mode provides an overall comparison of trip volume.
- These visuals help identify differences in mobility behavior between cities and transport modes.

2.4 Dashboard Design

- Consistent dark-blue visual styling is used for charts and KPI cards.
- The light-blue background provides a clear dashboard canvas.
- The layout moves from KPI summary to detailed mobility and weather analysis.

3.1 Trips by Weather Condition

The Trips by Weather Condition visual compares mobility activity under Storm, Rain, Fog, Cloudy and Clear conditions. It provides a simple categorical view of activity under different weather states.

3.2 Rainfall vs Passenger Demand

A clustered column chart was created to compare total passenger demand across rainfall categories.

Rainfall Category	Range	Meaning
Low	0–50 mm	Low rainfall
Moderate	51–100 mm	Moderate rainfall
High	101–150 mm	High rainfall
Very High	151+ mm	Very high rainfall

3.3 Calculated Columns

Rainfall Category was created using a SWITCH(TRUE()) calculated column based on Rainfall_MM. Rainfall Order assigns 1, 2, 3 and 4 to the four categories. Rainfall Category is sorted by Rainfall Order to display Low → Moderate → High → Very High.

3.4 Interpretation

- Passenger demand can be compared across increasing rainfall intensity.
- In this dataset, Very High rainfall has the highest total passenger demand among the four categories.

- Moderate and High rainfall categories show lower total passenger demand.
- These observations describe the supplied dataset and do not by themselves establish causation.

4.1 Milestone 3 Deliverables

Requirement	Implemented Visual / Feature	Status
Ride-hailing intelligence	Ride-Hailing Trips by City	Completed
Micromobility analytics	Micromobility Usage by City	Completed
Transport-mode analysis	Transport Mode Mix by City + Trips by Transport Mode	Completed
Weather sensitivity	Trips by Weather Condition	Completed
Demand vs rainfall	Rainfall vs Passenger Demand	Completed
Demand KPIs	Average Speed, Average Travel Time, Total Passengers, Average Traffic Density	Completed

Dashboard

This milestone extends the smart-city traffic dashboard into demand and mobility intelligence. The Power BI dashboard combines ride-hailing, transport-mode, micromobility and weather-related passenger analysis in one interactive report.



MILESTONE 4 – SMART URBAN MOBILITY DASHBOARD

Final Dashboard Development, Visualization and Performance Analysis

Objectives

- Develop a professional and interactive Power BI dashboard using the prepared mobility dataset.
- Present important mobility indicators through KPI cards for quick interpretation.
- Analyze monthly demand, traffic density and travel-time patterns.
- Compare mobility performance across cities and transport modes.
- Use latitude and longitude to visualize mobility demand geographically.
- Improve usability through slicers, consistent formatting and page navigation.
- Prepare a final dashboard suitable for demonstration and project evaluation.

1.1 Page 1 – Smart Urban Mobility Performance Overview

- KPI cards: Mobility Access Index, Average Reliability, Average Travel Time, Total Demand and Average Fare.
- Clustered bar chart: Mobility Access Index by City.
- Line chart: Total Demand by Month.
- Donut chart: Trips by Transport Mode.
- Table: City-wise Mobility Performance.
- Slicers: City and Transport_Mode.
- Navigation arrows for moving between dashboard sections.

1.2 Page 2 – Urban Traffic & Mobility Analytics

- Line chart: Average Traffic Density by Month.
- Line chart: Average Travel Time by Month.
- Pie chart: Average Fare by Transport Mode.
- Map: Urban Mobility Demand by Location & Mode.
- Navigation button for returning to the previous page.

Users can begin with KPI cards and city comparison, then move to detailed traffic, fare and geographic analysis. City and transport-mode slicers provide interactive filtering.

2.1 Dataset Understanding

The Smart Urban Mobility dataset contains trip, transport-mode, location, passenger, travel-time, traffic, weather and geographic information. It supports both time-series and spatial analysis.

2.2 Important Fields Used

Field / Measure	Purpose	Dashboard Usage
City	Geographic grouping	City comparison, table, slicer
Transport_Mode	Mode classification	Pie/donut charts, map legend, slicer
Month / Timestamp	Time dimension	Monthly trends
Total Demand	Overall demand	KPI, line chart, map bubble size
Mobility Access Index	Access indicator	KPI, city comparison, table
Reliability %	Reliability indicator	KPI and table
Average Travel Time	Travel-time indicator	KPI, line chart, table
Average Fare	Fare indicator	KPI, pie chart, table
Traffic_Density	Traffic condition	Monthly traffic trend
Latitude / Longitude	Geographic coordinates	Normal map

2.3 Data Cleaning & Transformation

- Unwanted columns were removed and relevant mobility fields were retained.
- City, transport, time, demand, fare, travel-time, traffic and coordinate fields were prepared for analysis.
- Null or incomplete values were reviewed before visualization.
- Date and month information was prepared for monthly trend analysis.
- Latitude and Longitude were configured as geographic fields and set to 'Don't summarize' for the map.
- Measures were created for KPI and visual aggregation requirements.

2.4 Data Quality Considerations

Aggregation settings are important for correct reporting. Latitude and Longitude must represent coordinate pairs, while measures such as Total Demand and Average Fare are aggregated according to their analytical purpose.

3.1 Role of DAX Measures

DAX measures provide reusable calculations that respond to slicers and filters. They ensure that demand, fare, reliability, travel-time and mobility-access calculations remain consistent across the dashboard.

3.2 Major Measures

Measure	Purpose	Used In
Mobility Access Index	Mobility-access indicator	KPI, city chart, table
Average Reliability	Overall reliability level	KPI, table
Average Travel Time	Average travel duration	KPI, monthly trend, table
Total Demand	Overall mobility demand	KPI, monthly trend, map
Average Fare	Average fare value	KPI, pie chart, table
Total Trips	Trip count	Transport-mode chart
Average Traffic Density	Monthly traffic-density value	Traffic trend

3.3 KPI Cards

KPI	Displayed Value
Mobility Access Index	48.64
Average Reliability	2.63%
Average Travel Time	90
Total Demand	2415
Average Fare	109.93

3.4 Benefits of KPI Design

- Provides an immediate summary of the current mobility condition.
- Allows quick comparison when slicers are applied.
- Creates a clear starting point before detailed visual analysis.
- Makes the dashboard suitable for project presentation and evaluation.

4.1 Page 1 Visuals

- Mobility Access Index by City – clustered bar chart for city comparison.
- Total Demand by Month – line chart for monthly demand variation.
- Trips by Transport Mode – donut chart for transport-mode distribution.
- City-wise Mobility Performance – table combining access, reliability, travel time, fare and demand.

4.2 Page 2 Visuals

- Average Traffic Density by Month – monthly traffic-density trend.
- Average Travel Time by Month – monthly travel-time trend.
- Average Fare by Transport Mode – pie chart comparing average fare values.

- Urban Mobility Demand by Location & Mode – map using Latitude and Longitude, demand-sized bubbles and transport-mode categories.

4.3 Slicers and Filtering

- City slicer – filters the dashboard by geographic location.
- Transport_Mode slicer – filters the report by transport mode.
- Dropdown style keeps the dashboard compact and easy to use.

4.4 Geographic Map

The final dashboard uses a normal Power BI Map. Latitude and Longitude are set to 'Don't summarize'. Total Demand controls bubble size and Transport_Mode provides the legend, allowing geographic demand and mode distribution to be viewed together.

4.5 Page Navigation

Navigation buttons were added to support movement between dashboard pages. The button action is configured as Page navigation with the required destination. In Power BI Desktop editing mode, navigation is tested with Ctrl + Click; in the published/viewing experience, the navigation button can be used normally.



Figure 1. Milestone 4 mobility performance dashboard

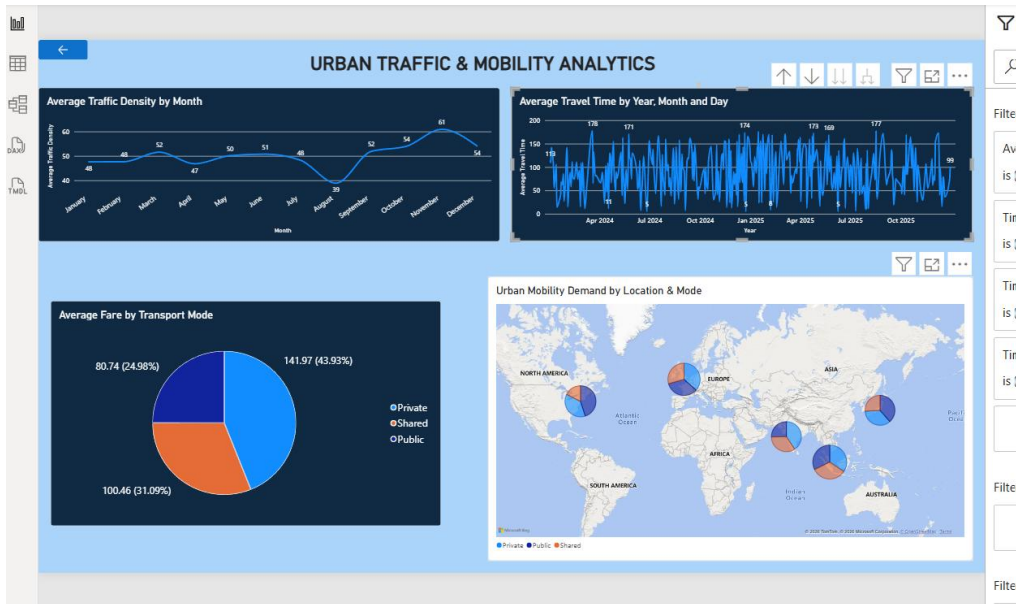


Figure 2. Milestone 4 traffic and mobility analytics dashboard

5.1 Final Outcome

Milestone 4 completes the transition from prepared mobility data to an interactive Power BI analytical dashboard. The final report combines numerical KPIs, time-series trends, transport-mode comparisons, city-level performance, geographic visualization, filtering and navigation.

CONCLUSION

The internship project, Smart Urban Mobility and Traffic Intelligence, was completed through four milestones using Microsoft Power BI. The project progressed from dataset understanding and data preparation to interactive dashboard development, demand intelligence, and advanced mobility performance analysis. By implementing Power Query, DAX measures, KPI cards, slicers, charts, geographical maps, and page navigation, the project transformed raw transportation data into meaningful visual insights. Overall, the internship enhanced my practical skills in data analytics, data visualization, business intelligence, and problem-solving, while demonstrating how data-driven insights can support better urban transportation planning and decision-making.

